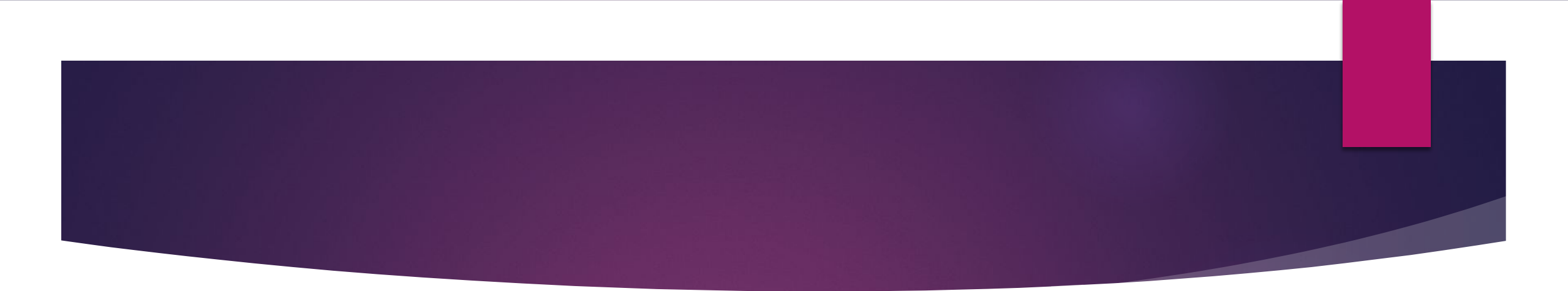


Information Systems, Organization and Strategy

-NILAM SANGROULA

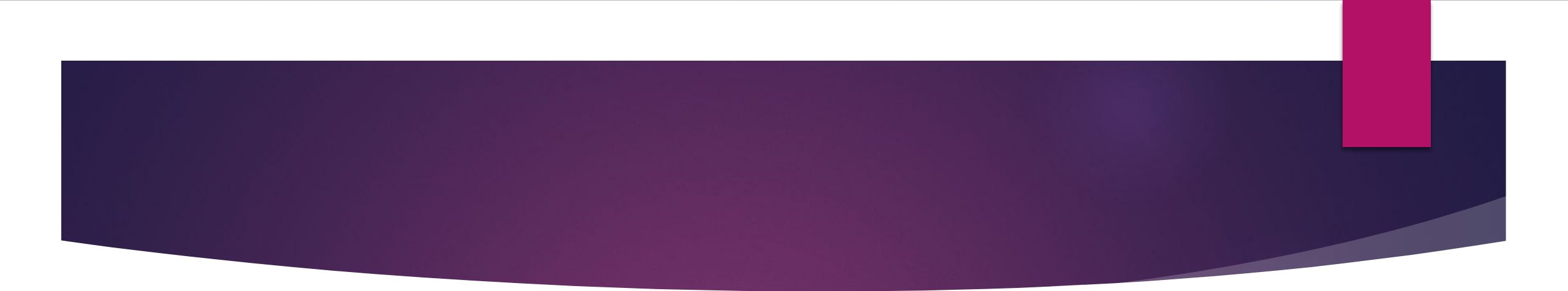
Organizations and Information Systems: Definition

- ▶ An organization is a body built for a collection of individuals who join together to achieve some common goals and objectives bounded by legal entities. Organizations are often referred to as a company, institution, association, government body, etc. They follow certain legal procedures like business registration, tax identification, and maintaining corporate book records.
- ▶ An organization is a stable, formal social structure that takes resources from the environment (Capital and labor) and processes them to produce outputs.
- ▶ The organization (the firm) transforms these inputs into products and services in a production function. The products and services are consumed by environments in return for supply inputs.

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- ▶ The organization influences the design, implementation, and use of information systems, while the information system, in turn, affects the structure, processes, and operations of the organization. In this blog, we will explore the two-way relationship between an organization and its information system.

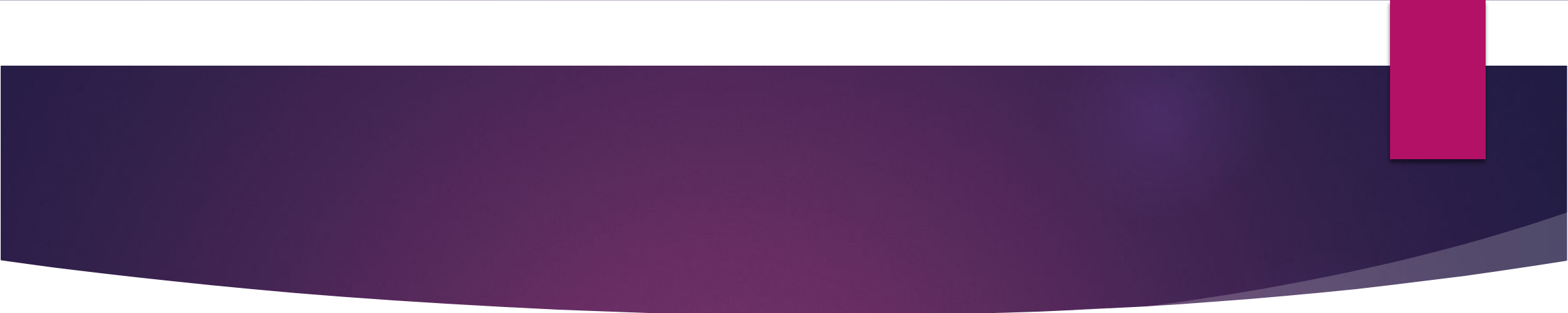
Features of Organizations:

- ▶ Clear division of labor
- ▶ Hierarchy of authority
- ▶ Abstract rules and procedures
- ▶ Impartial judgments
- ▶ Technical qualifications for positions
- ▶ Maximum organizational efficiency
- ▶ Routines and Business Processes
- ▶ Organization Politics
- ▶ Organization Culture
- ▶ Organization Environments
- ▶ Organizational Structure



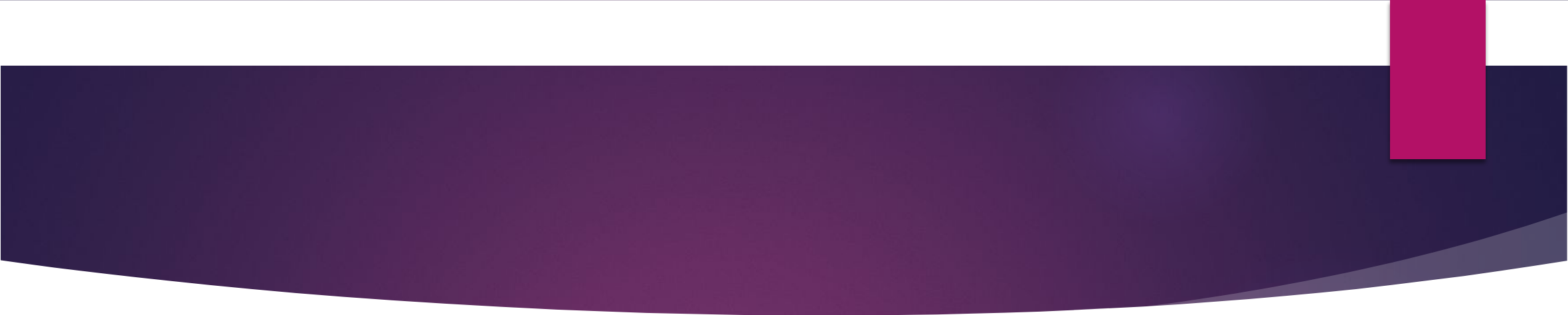
► **Business processes**

- Simply, the business processes are the tasks and behaviors used by an organization in order to achieve desired goals and objectives. Business processes are the flows of information, materials, and knowledge in the organization. So an organization can be identified as a collection of the business process. Mainly there are some common business processes which include accounting & finance processes, manufacturing processes, marketing processes, and human resources.
- When an organization operating its business processes, the organization needs to engage with various information. Stakeholders of the business provide much information to each business process; in this case, information systems help to manage the business process well to ensure organizational performance.



► **Organizational politics**

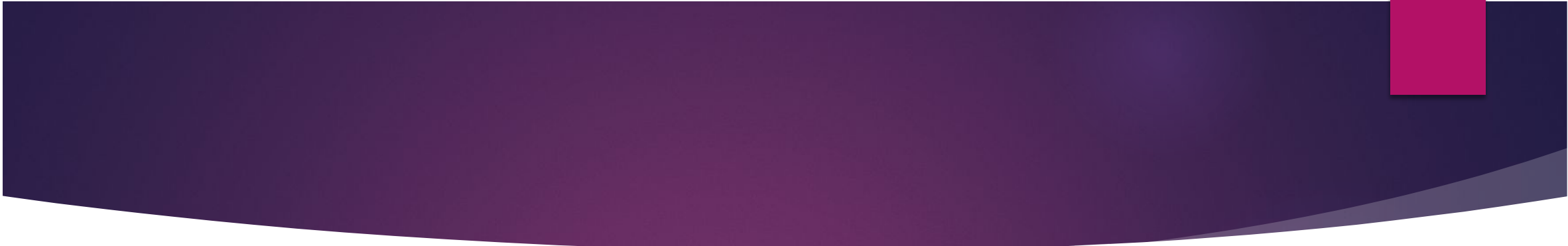
- Basically, organizational politics refers to behaviors or attitudes that influence positively or negatively on others. Organizational politics help to develop interpersonal relationships in the organization and increase organizational performance. The people who work in the organization operate their jobs under different job positions with different specialties.
- So all of them have different points of view about how resources and other things should be distributed. These differences are so important to managers as well as employees in the organization. Because of these differences, the organization may have to face struggles for competition, Resources, and conflict.
- The political resistance can be identified as the great difficulty among the rest of the difficulties- especially when the organization invests in information systems, it will make a big change in business objectives, business processes, and strategies. In this case, the people who know how to work with organizational politics will become successful in the organization rather than less-skilled people.



► **Organizational culture**

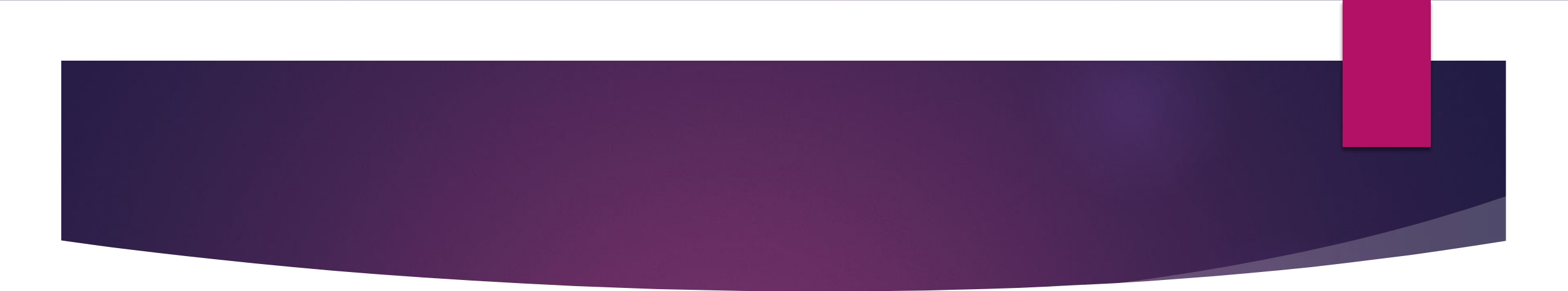
- Organizational culture includes beliefs, assumptions, an organization's expectations, and values that interact with the social and psychological environment of the entity.

Organizational culture or corporate culture covers a set of assumptions about what to produce, how to produce, where should produce, and for whom to produce. So this is a powerful element for the organization. By reason of this culture, the organization highly consider technological change and the impacts on the organization's culture.



► **Organizational environment**

- To operate business processes smoothly, the organizational environment is a necessity for the organization. Every organization is depending on and opens to its physical and social environment. So, both organization and environment have a reciprocal relationship.
- Basically, the environment changes faster than organizations. These changes make strengths, weaknesses, opportunities, and threats to the organization. Therefore every organization does an environmental analysis before operating their business processes.
- In this case, information systems play a vital role as an instrument for the organization. Information systems provide information about their customers, suppliers, and other related information to help managers in the decision-making process. Also, information systems help managers to identify environmental changes that the organization needs to respond to.



► **Organizational structure**

- Every organization in the world, have an organizational structure or shape. According to Mintzberg's classification, there are five basic kinds of organizational structures which include entrepreneurial structure, machine bureaucracy, divisional sized bureaucracy, professional bureaucracy, and adhocracy. The structure of an organization is so important for information systems. A well-designed structure is very useful for the flow of information among the organization and it is very easy to make improved decisions for each management level.

Organization's Influence on Information Systems

- ▶ **Information Needs:** The organization's goals, strategies, and operations determine the information needs. These needs shape the design and functionality of the information system. For example, an organization that emphasizes data-driven decision-making will require an information system capable of capturing and analyzing relevant data.
- ▶ **Organizational Structure:** The structure of an organization influences the information system's design. The hierarchy, departments, and reporting relationships within the organization determine the flow of information and the level of integration required in the information system.
- ▶ **Business Processes:** The organization's business processes and workflows shape the information system's functionalities. The system is designed to support and automate these processes, ensuring smooth operations and efficient data flow.
- ▶ **Organizational Culture:** The organizational culture, including values, norms, and beliefs, impacts how the information system is perceived, adopted, and used by employees. A culture that values innovation and technology adoption will likely embrace the information system more readily.

Information System's Impact on the Organization

- ▶ **Efficiency and Effectiveness:** A well-designed and implemented information system improves the organization's efficiency and effectiveness by automating manual tasks, streamlining processes, and facilitating data-driven decision-making. This, in turn, enhances productivity, reduces errors, and improves overall performance.
- ▶ **Communication and Collaboration:** Information systems promote communication and collaboration within the organization. They provide platforms for sharing information, exchanging ideas, and working on projects collectively, fostering a culture of collaboration and knowledge-sharing.
- ▶ **Decision-Making Processes:** Information systems provide accurate and timely data, analysis, and reports, enabling managers to make informed decisions. They support different levels of decision-making, from operational to strategic, by providing the necessary information and tools for analysis.
- ▶ **Organizational Learning:** Information systems facilitate organizational learning by capturing and storing knowledge and best practices. They enable the organization to build a knowledge repository, share lessons learned, and leverage past experiences for future decision-making and innovation.

► **Thus,**

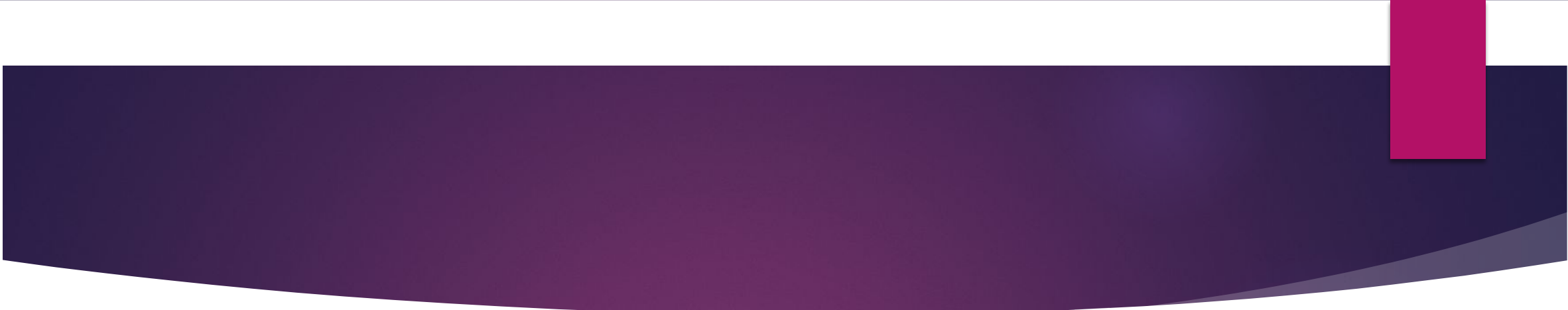
- The relationship between an organization and its information system is not static. It requires continuous interaction, feedback, and adaptation. As the organization evolves, the information system needs to be updated, modified, and aligned with changing requirements and technologies. Similarly, the organization needs to adapt its processes, structures, and culture to fully leverage the capabilities of the information system.

Using Information Systems to Achieve Competitive Advantage:

- ▶ In almost every industry you examine, you will find that some firms do better than most others. There's almost always a stand-out firm.
 - ▶ In the automotive industry, Toyota is considered a superior performer.
 - ▶ In pure online retail, Amazon is the leader.
 - ▶ In off-line retail, Walmart is the largest retailer on earth, is the leader.
 - ▶ In online music, Apple's iTunes is considered the leader with more than 75 percent of the downloaded music market,
 - ▶ In the industry of digital music players, the iPod is the leader.
 - ▶ In Web search, Google is considered the leader.
- ▶ Firms that “do better” than others are said to have a competitive advantage over others: They either have access to special resources that others do not, or they are able to use commonly available resources more efficiently—usually because of superior knowledge and information assets.

Information systems can be used to gain a competitive advantage in various ways

- ▶ **Improved Efficiency:** Information systems can automate and streamline many business processes, improving efficiency and reducing costs. This means that organizations can produce goods and services at a lower cost than their competitors, which can be a significant competitive advantage.
- ▶ **Enhanced Customer Service:** Information systems enable organizations to collect and analyze data about their customers, allowing them to provide more personalized and responsive customer service. This can lead to increased customer loyalty and retention, which can be a key competitive advantage in many industries.

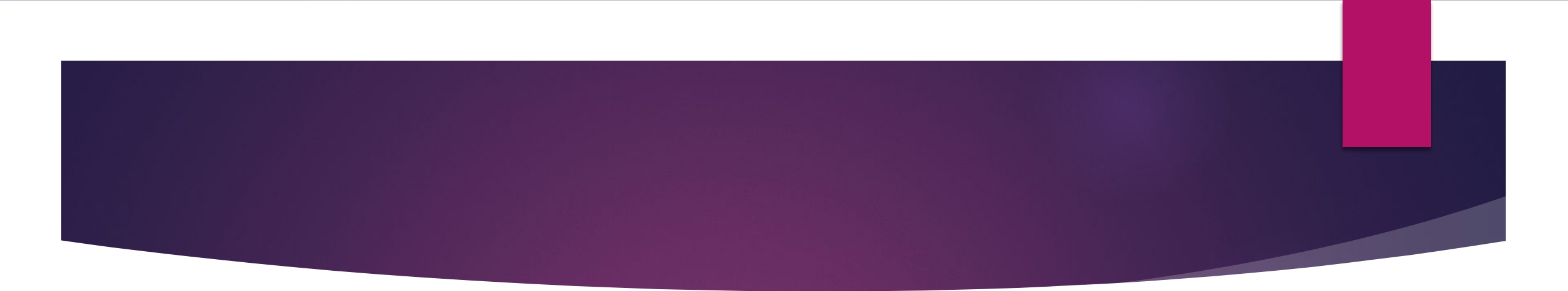
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- ▶ **Product Innovation:** Information systems can help organizations to innovate by providing insights into customer preferences, market trends, and emerging technologies. This can help organizations to develop new products and services that meet the changing needs of their customers, giving them a competitive advantage.
 - ▶ **Faster Decision-Making:** Information systems provide real-time data, analytics, and reporting, enabling organizations to make faster and more informed decisions. This can help organizations to respond quickly to market changes and stay ahead of their competitors.

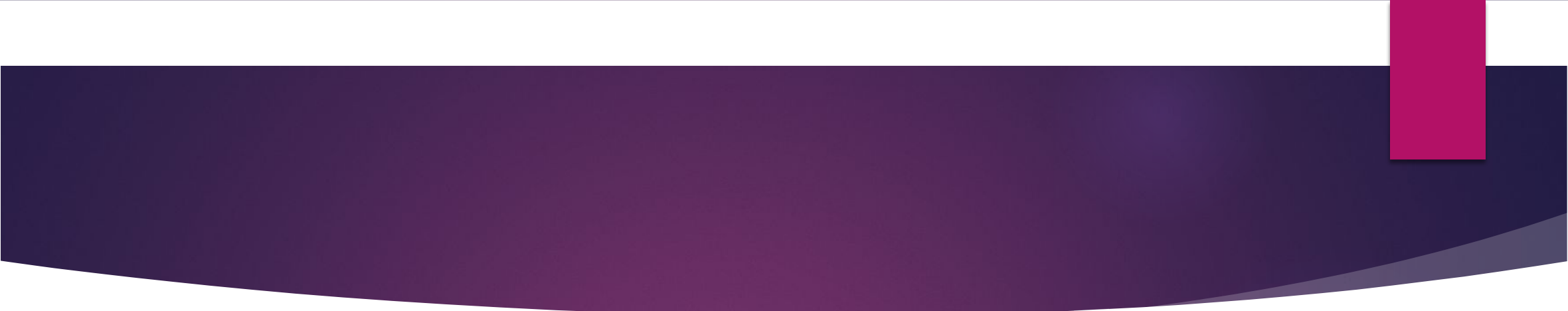
MICHAEL PORTER'S COMPETITIVE ADVANTAGE MODELS

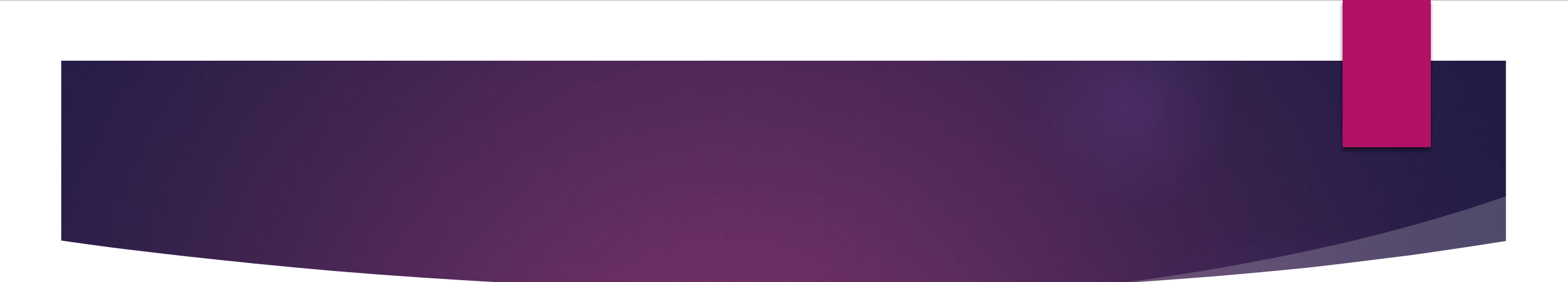
- ▶ Michael Porter's value chain model is a framework that identifies the primary activities and support activities that are required to create value for customers. The primary activities include inbound logistics, operations, outbound logistics, marketing and sales, and service, while the support activities include procurement, technology development, human resource management, and firm infrastructure.
- ▶ By analyzing each of these activities and identifying how they can be streamlined or enhanced, a company can create a competitive advantage.

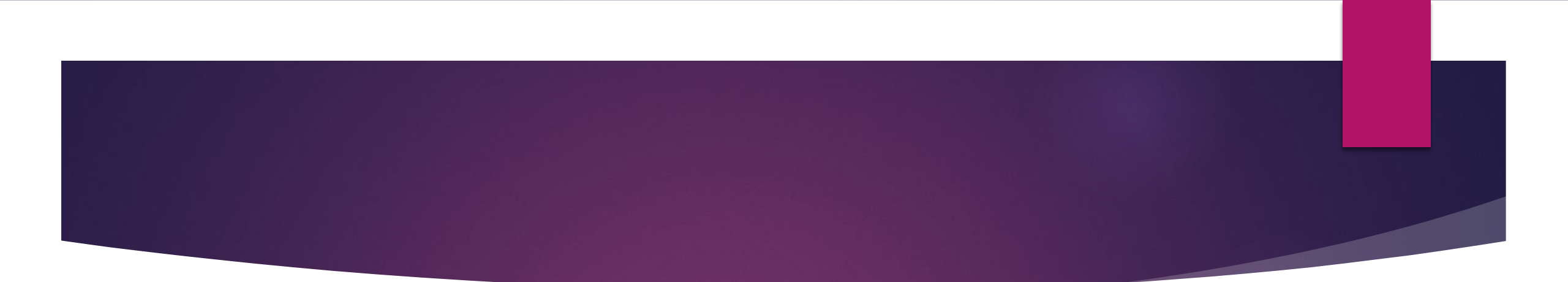


Porter's Value Chain Model

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- ▶ **Inbound Logistics:** A company might use information systems to track and manage the delivery of raw materials and supplies from suppliers. This can include using barcode scanning technology to track inventory levels, using automated replenishment systems to ensure that supplies are ordered and delivered on time, and using data analytics to forecast demand and optimize inventory levels.
 - ▶ **Operations:** A company might use information systems to automate and streamline manufacturing processes. This can include using robotics and automation technologies to increase efficiency and reduce waste, using advanced analytics to optimize production schedules, and using quality control systems to monitor and improve product quality.

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- ▶ **Outbound Logistics:** A company might use information systems to manage the distribution of products to customers. This can include using logistics software to optimize shipping routes and reduce delivery times, using real-time tracking systems to monitor shipments, and using customer relationship management (CRM) tools to manage customer accounts and preferences.
 - ▶ **Marketing and Sales:** A company might use information systems to target and engage customers. This can include using digital marketing platforms to reach customers through social media, email, and other channels, using data analytics to personalize marketing messages and offers, and using customer feedback systems to collect and respond to customer reviews and complaints.

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 - ▶ **Service:** A company might use information systems to provide customer service and support. This can include using chatbots and other AI-based technologies to provide quick and efficient support, using customer satisfaction surveys to measure and improve service quality, and using customer relationship management (CRM) tools to manage customer accounts and preferences.

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- ▶ **Procurement:** A company might use information systems to manage the procurement of goods and services from suppliers. This can include using online procurement systems to manage the bidding and purchasing process, using data analytics to forecast demand and optimize inventory levels, and using supplier relationship management (SRM) tools to monitor and manage supplier performance.
 - ▶ **Technology Development:** A company might use information systems to support research and development (R&D) efforts. This can include using data analytics and machine learning to identify new product opportunities, using simulation and modeling tools to test and optimize product designs, and using collaborative platforms to facilitate cross-functional teamwork and innovation.
 - ▶ **Human Resource Management:** A company might use information systems to manage its workforce. This can include using HR management software to track employee performance, manage compensation and benefits, and support employee development and training programs.